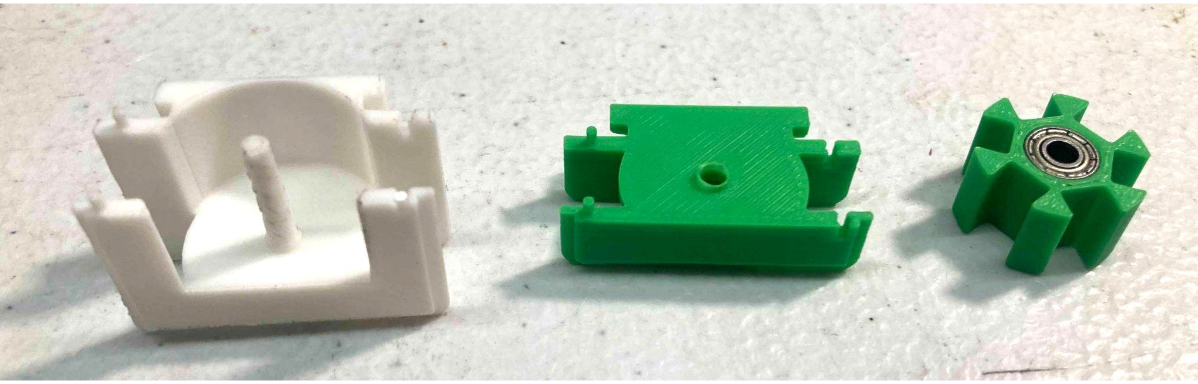
put bearing+ring+ bearing sandwich into the magnet holder

one at a time!

no magnets in here for a while... :-(

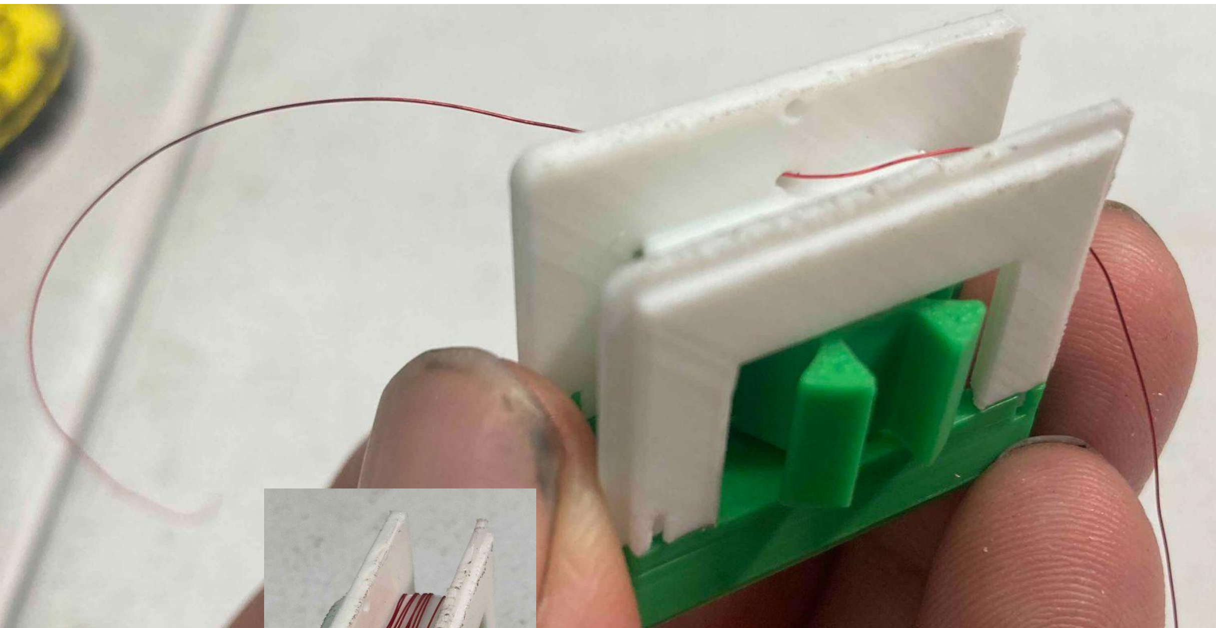
magnet ring assembly

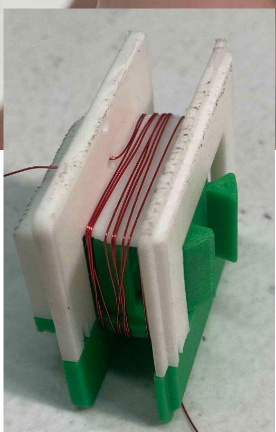
1. put magnet ring onto bottom bobbin shaft
2. add top bobbin



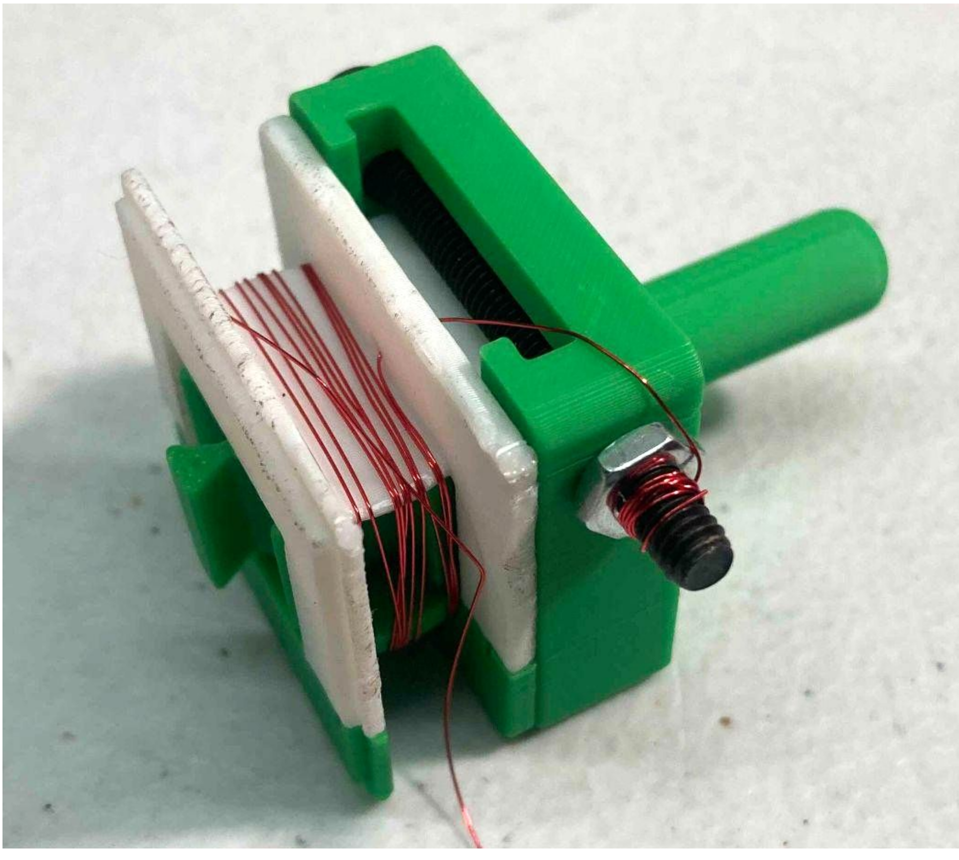
Wire Prep

1. **feed magnet wire through inside hole in bobbin (notice the hole above it too!)**
2. **pull ~5-6” (15cm) through**
3. **do about 20 turns by hand.**
   a. make the loops tight, not loose!





1. **slide this assembly into the winding attachment.**
2. **put in the M4 screw and nut (hand tight) to lock.**
3. **wrap the excess 6” of wire around the end of the screw (so it is not in the way)**

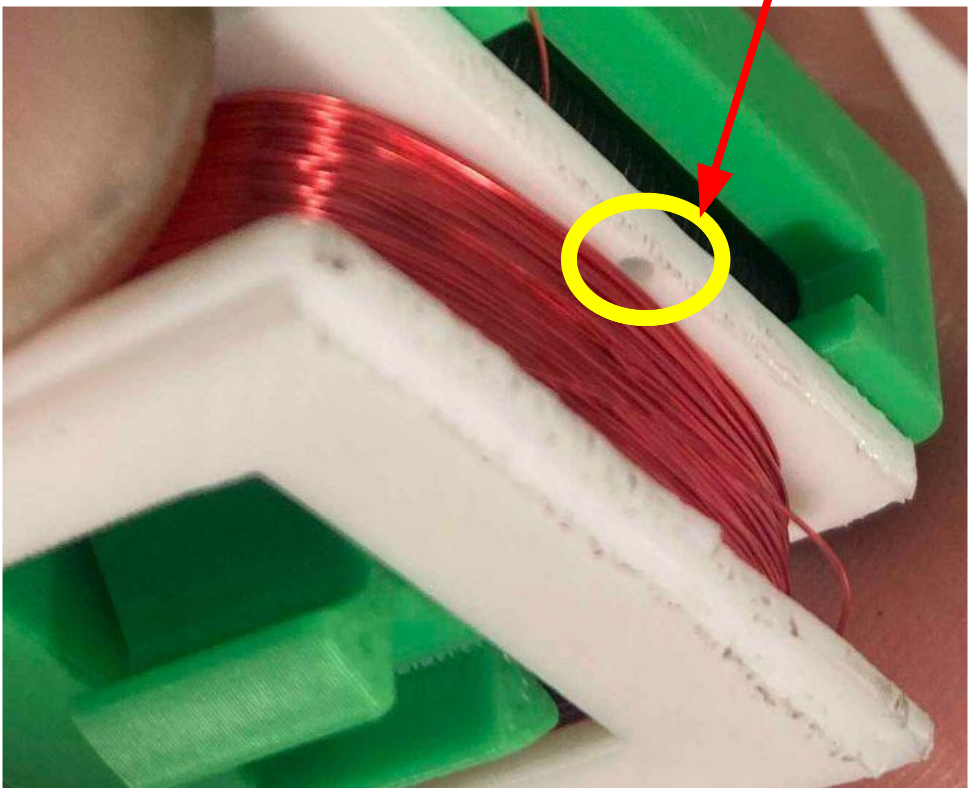


Coil Pre-Wind!

1. **partner up!**
2. **load the previous assembly into a drill chuck**
3. **have your friend hold the magnet wire spool**
4. **hold drill in one hand, and pinch the wire between your fingers in the other hand (or your friend can hold spool and wire, and you hold drill)**
    a. we will need add **tension** to the wire and guiding it onto the bobbin evenly. we want **tight even loops**
    b. **don't pinch TOO tight, you may break the wire and then you have to rewind**
    c. you can use a piece of plastic or paper between your fingers and the wire if holding it with bare hands is uncomfortable

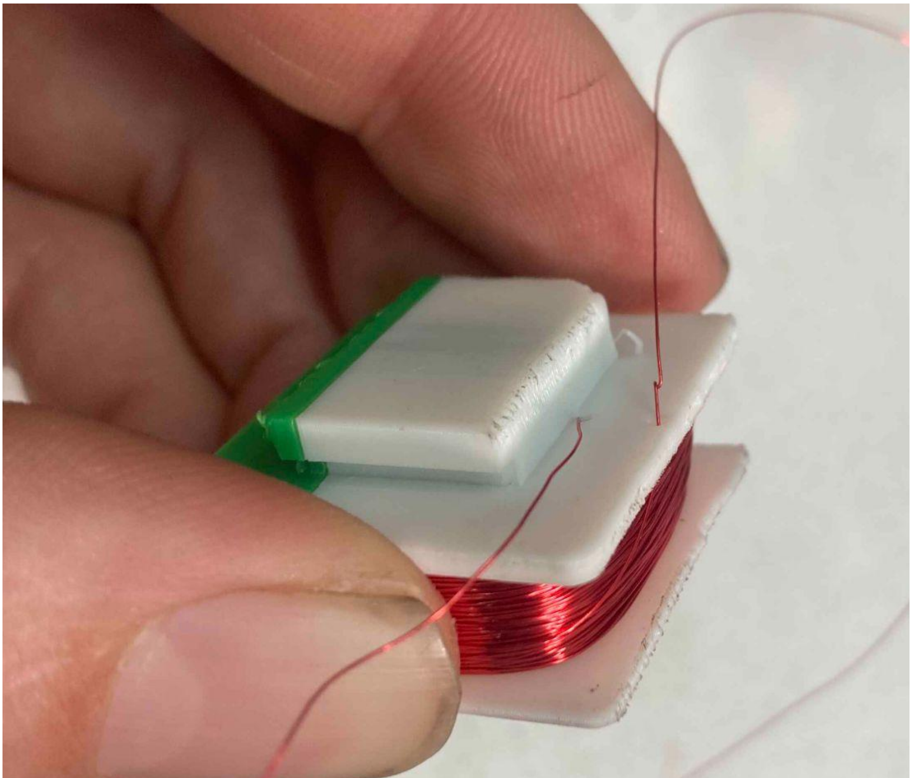
Coil Winding!

1. **once you have everything ready, start spinning the bobbin. make sure you are spinning inthe right direction! go verrrrry slow at first!**
    a. The goal is **tight** windings, evenly distributed across the bobbin.
2. **Guide the wire left to right to left slowly as the bobbin rotates.**
3. **Wind until spool is empty OR you are just below the outer hole in the bobbin. (above where you fed in the initial wire).**
4. **stop and cut the wire about 12" long**



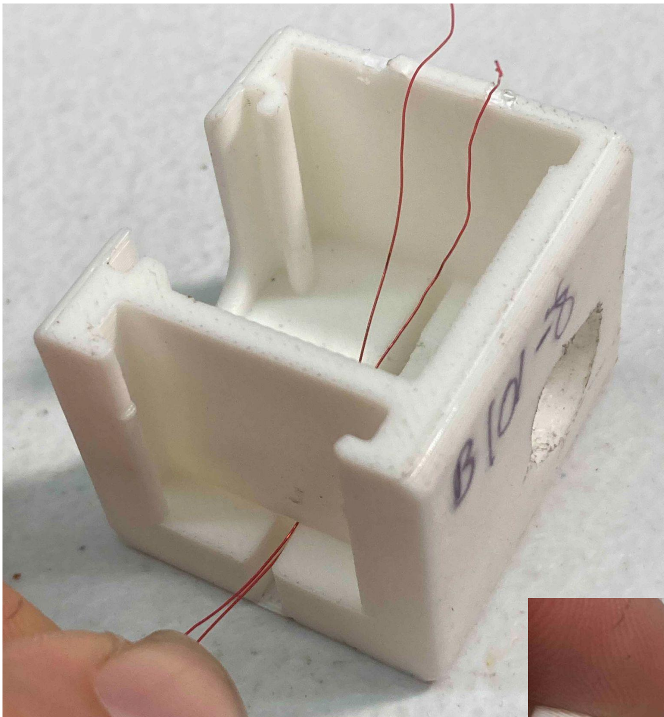
remove bobbin from attachment

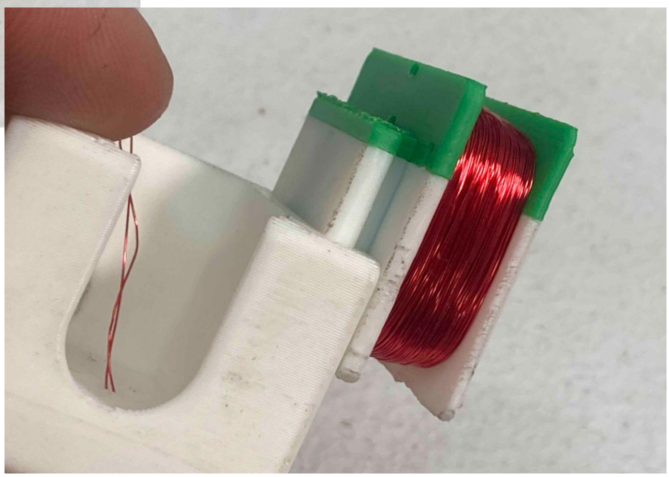
1. unwind excess wire from screw
2. take out nut and screw
3. remove bobbin
4. wind the wire around the bobbin 6 or so times tightly.
5. feed the wire through the upper hole



Coil Installation

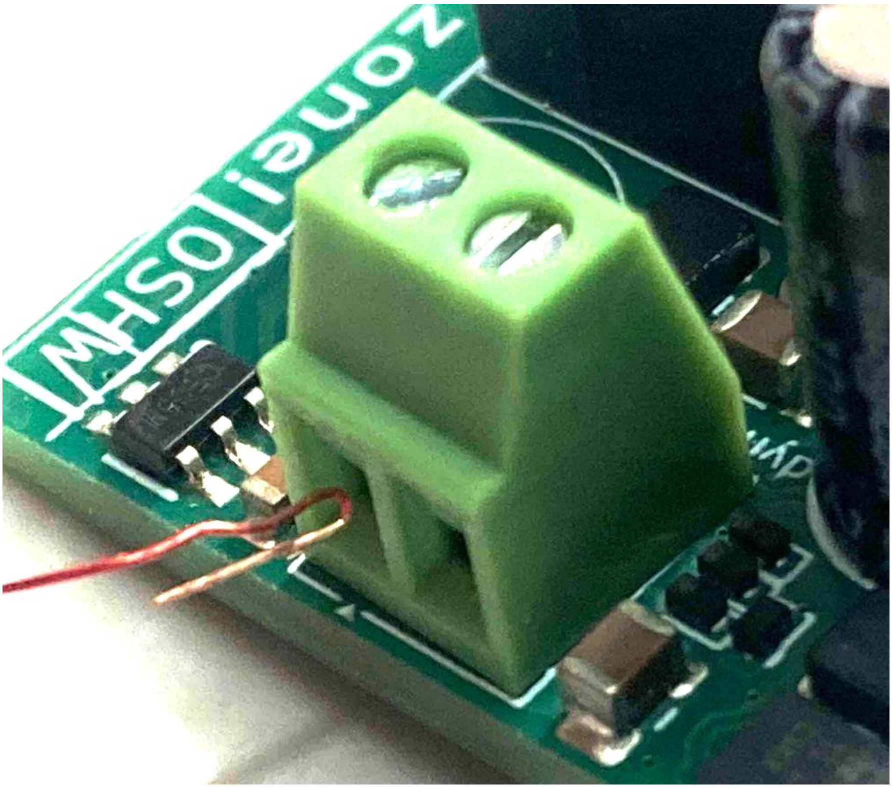
1. **cut both wires down to ~ 3” (7 cm).**
2. **feed wire through the slot in bottom of the main enclosure**
3. **slide bobbin onto enclosure with wires coming out bottom**





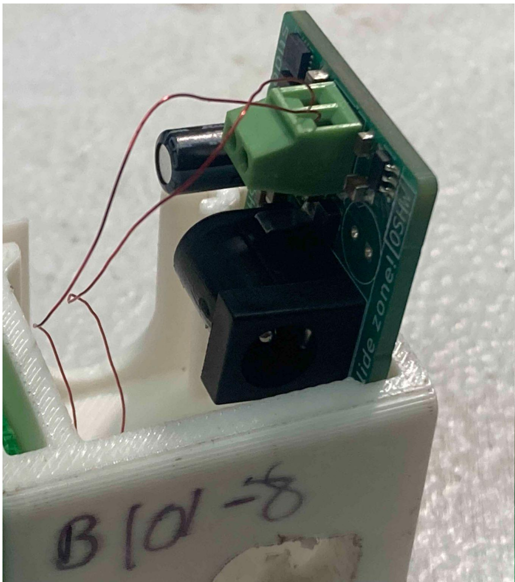
COIL PCB wiring

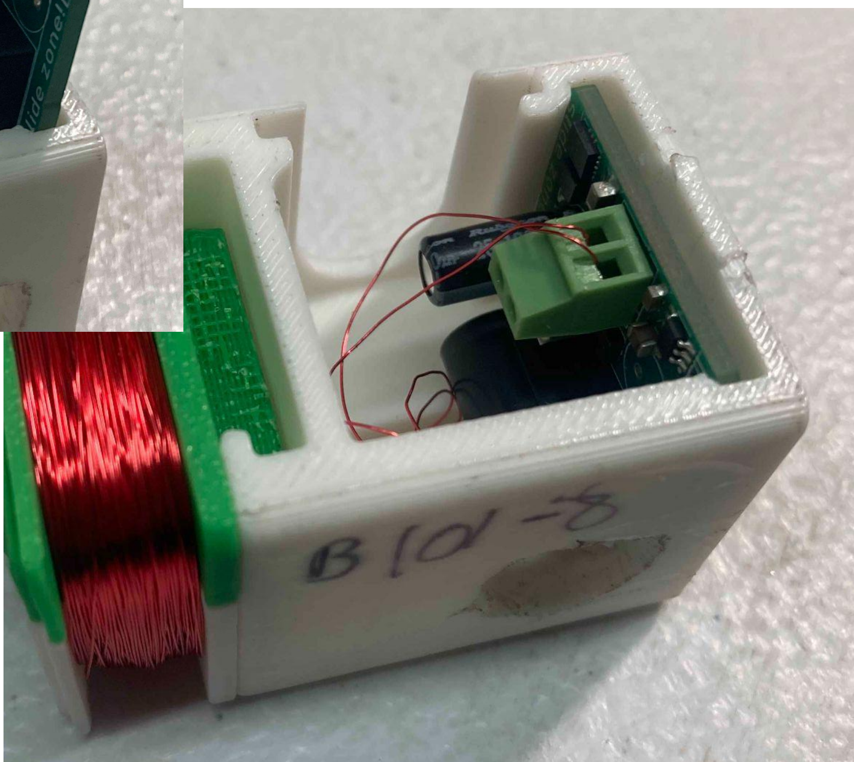
1. use sandpaper to sand off about ¼" off the ends of the wires.
2. the color should wipe off and you see bare copper.
3. Loosen the small screw (using the small flathead screwdriver) on the green connector on the rectifier PCB.
4. We will be putting the wires into this connector.
5. fold about ⅛" back of each wire. (making a shepherds hook-type situation)
6. put the wire into one of the small holes on the side of the green connector. tighten with screwdriver.
7. Add the other wire to the remaining hole on the connector. Tighten with screwdriver.



PCB Installation

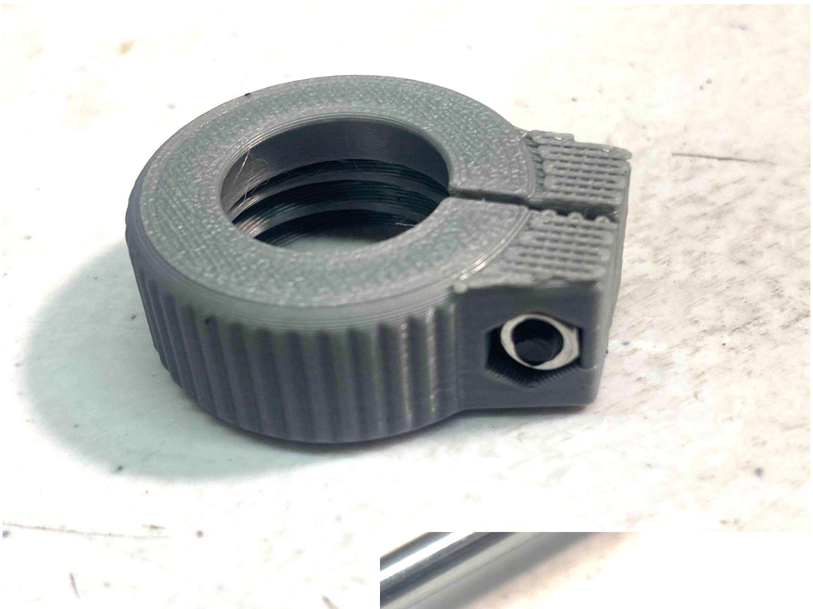
1. Side PCB into the rear of the enclosure, making sure the green connector is facing up!

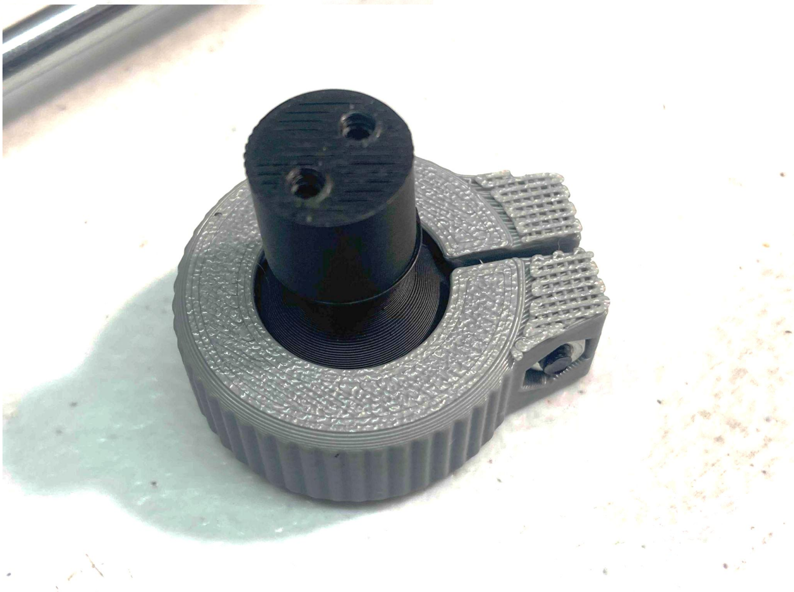




Assemble bike mount! Pt. 1

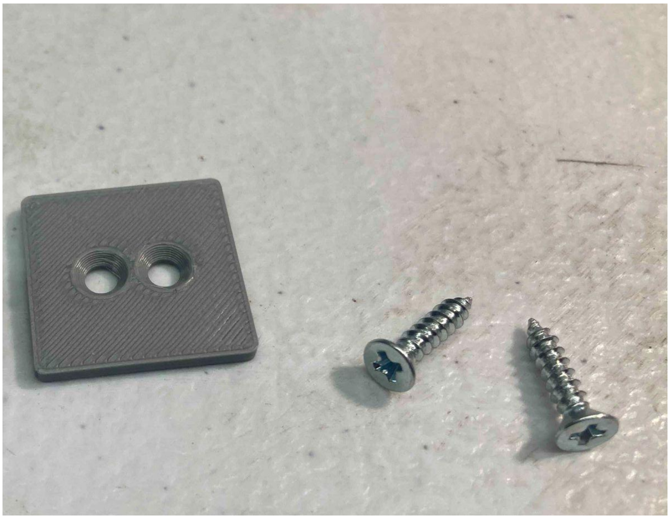
1. Add M3x16 screw and M3 nut on the large 3D printed nut. Loosely tighten. This is to be tightened all the way when the nut is on the actual bicycle mount on the bike
2. Put the ball down on the table with shaft up
3. Place the 3D printed nut around the shaft with threads facing down
4. If it's upside down, you can see the threads.

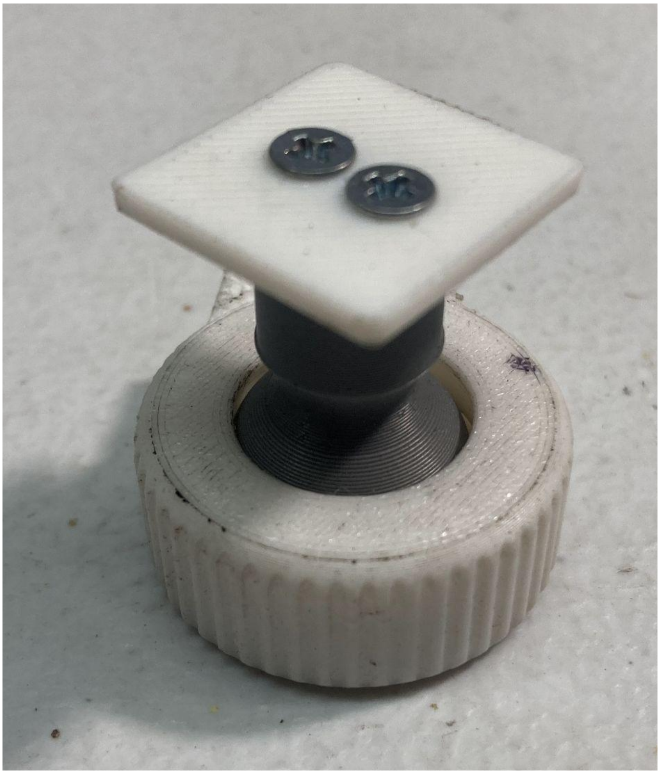




Assemble bike mount! Pt 2

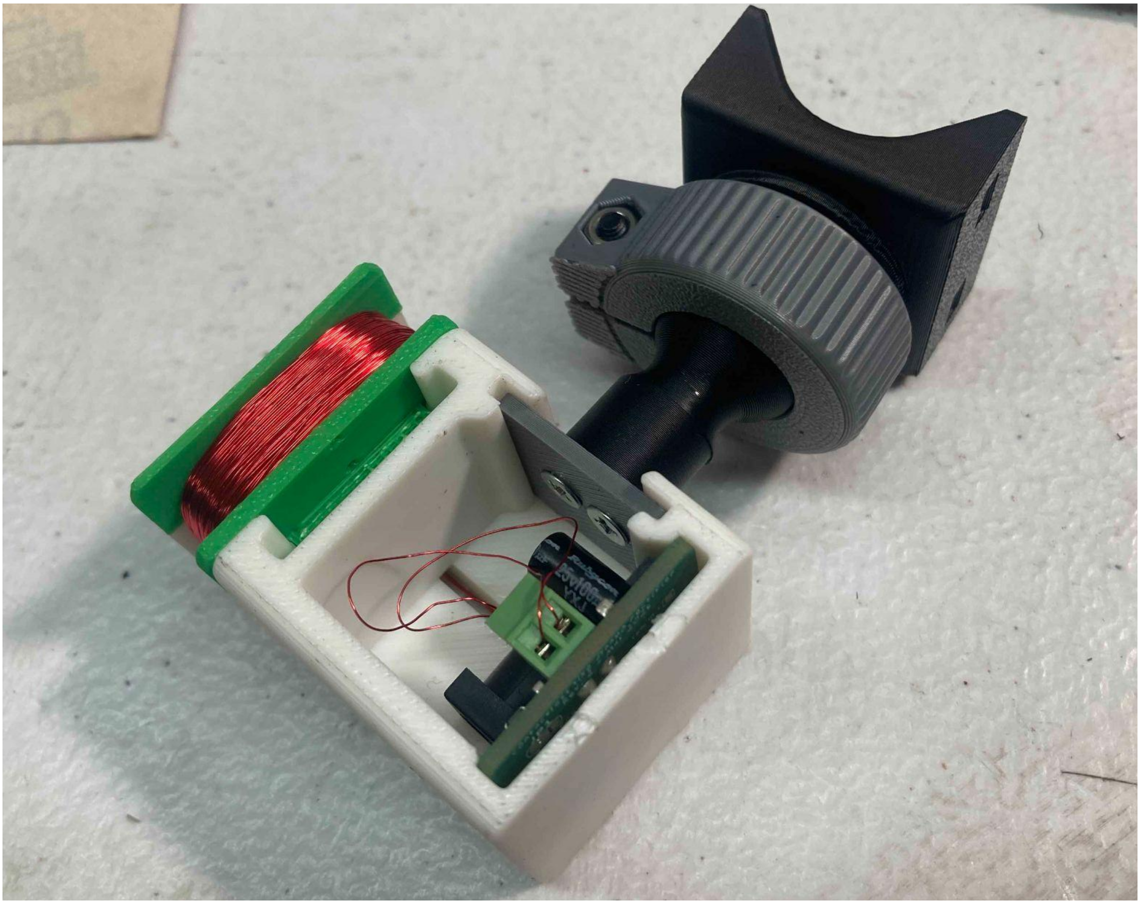
1. **Add the plate and 2 screws.**
2. **Make sure the screws go through the side of the plate that is countersunk.**





Assemble bike mount! Pt 3

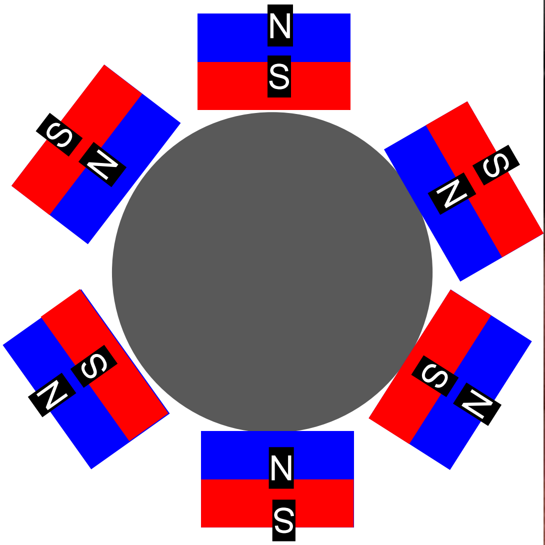
1. Screw in the ball nut into the bike mount.
2. slide this assembly onto the enclosure
3. Add Cover

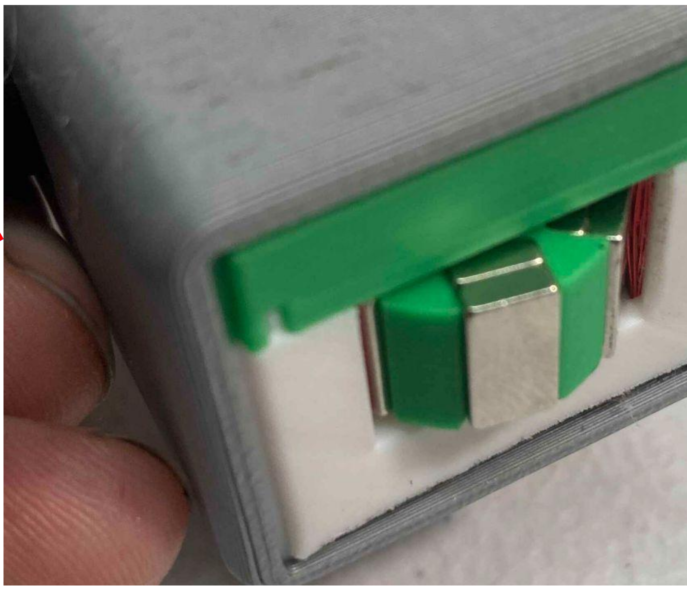


Add Magnets to Magnet Ring!

caution! These are very strong!

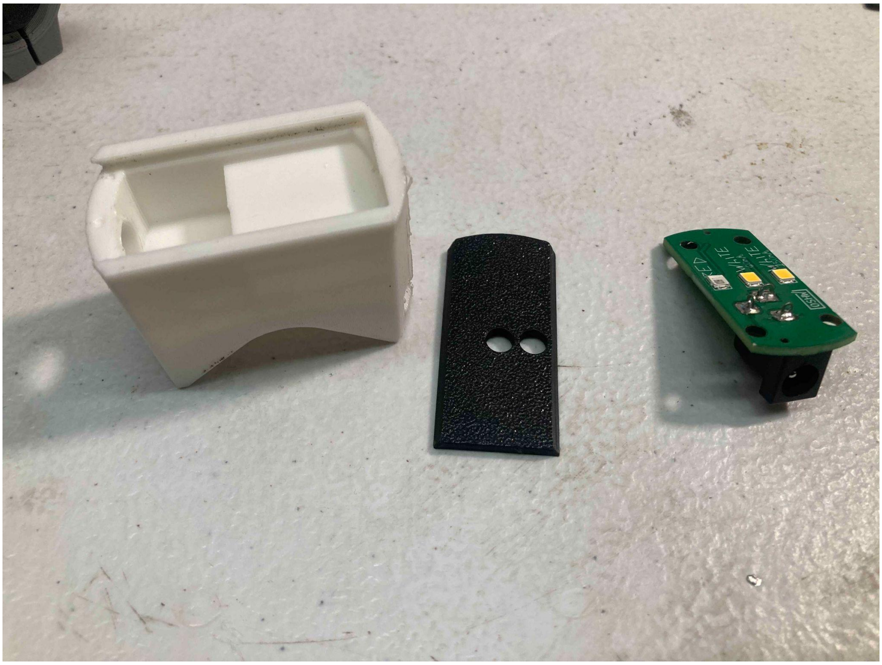
1. **Pull off 2 magnets from the magnet stack**
2. **Put these two into a slot on the magnet ring**
3. **Pull off 2 more magnets**
4. **Flip these and put in the slot next to the first two.**
    a. test by holding the pair close to the previous one and seeing if they attract or repel.
    b. if they repel that is good, put them into the next slot that way. if not, flip them
5. **do this for the remaining slots**





Assemble Light!

1. **Add Light PCB to the enclosure.**
2. **Stick the lens into the small holes on the lid.**
3. **Slide the lid onto the case.**



Mounting!

1. Find a good place for the dynamo to live on a fork.
2. The magnets need to be about 3mm - 5mm away from the rim.
3. Add some bike inner tube to that area and tape it down.
4. Mount the dynamo with zip ties
5. Mount light somewhere on the bike with zip ties!
6. Connect the two with the 3' cable.
7. Secure cable with zip ties.
8. LIGHT UP THE NIGHT WITHOUT SOME FREE ENERGY! AND NO DANG BATTERIES !

this has been a
freakylamps.com
production
 - w. kennedy

contactless bicycle dynamo workshop!

Make a bicycle light without a battery!